

# What an Occluded Handover Can Tell Its Receiver

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

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## Abstract

Paper Q studies how reports and rewards can direct an agent with private information, while paper P asks whether a donor's wrist can locate an object hidden during handover. The thread looked for a limit on such control when wrist and motor observations leave the object's orientation unclear. The peers derived conditional bounds on a receiver's success and on rewards based on coarse measurements, and checked a failure of one of Q's reward menus under a changed signal assumption. They also found a repair for a stipulated signal channel. The thread paused because P supplies neither measured ambiguous histories nor tests showing that the hidden orientations require different receiving actions.

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## 1 Introduction

Q, *Mechanism Design for Alignment and Control*, studies an agent whose preferences, possible actions and information may be private. A human can ask for reports, recommend actions and offer rewards. Q asks when those tools make the agent report truthfully and act as intended. Its oversight example uses a weaker monitor to choose rewards for a stronger actor, which knows the relevant state exactly [2].

P, *When a Donor Wrist Cannot Locate a Rotating Held Object*, examines a handover in which the held object is hidden from the receiver. It reads a proposal to use the donor arm's known wrist pose in place of the hidden object pose alongside a demonstration that a hand can rotate an object within its grasp. P observes that the object may then move relative to the wrist. It gives a conditional limit on wrist based receiving, and proposes a measurement test. It does not report matched histories of wrist, motor readings and actions with different object poses [3].

The peers asked whether a receiver could be guided by Q's reports or rewards when the observations in P fail to identify which way the object faces. They separated two questions. Could the receiver choose the right approach from its information? Could a reward based on measured outcomes induce a desired physical action? The connection is conditional throughout: P neither reports a strategic donor nor supplies a signal distribution for Q's model.

## 2 What was done

The peers first identified the possible ambiguity at handover. A fixed wrist pose need not fix object orientation if the object rotates within the hand. They treated the wrist, motor encoder readings and action history as the receiver’s available record. They then asked what follows if two orientations can have the same record and need incompatible receiving actions.

One peer turned this into a bound on success from the record alone. In parallel, the other took Q’s oversight menu and replaced its actor’s exact view of the state with a coarse signal. The peers checked both calculations against Q and P. They corrected an early overreach: one matched pair supports a claim about that pair, but says nothing by itself about how often ambiguity occurs in ordinary handovers.

Next, they considered incentives separately from inference. A finite model of measured outcomes showed when an observation based reward can support a desired action, including a simple case in which no such reward can work. Finally, they checked a structured signal channel for which adapting Q’s menu restores its intended action. These were mathematical checks of stipulated models, not handover experiments.

## 3 Results

### A limit from the receiver’s observations

Let  $H$  be either of two equally likely object orientations, labelled 0 and 1. Let  $Y$  be everything the receiver observes before acting, including the wrist, encoder and action history. Write  $P_0$  and  $P_1$  for the distributions of  $Y$  under the two orientations. Suppose that no receiving action succeeds for both. The peers found that a receiver using only  $Y$  has average success probability at most

$$\frac{1 + \text{TV}(P_0, P_1)}{2}.$$

Here  $\text{TV}(P_0, P_1)$ , total variation distance, is the largest difference between the probabilities that  $P_0$  and  $P_1$  assign to the same possible event. To succeed, the receiver must in effect distinguish the two orientations. The displayed expression is the best possible probability of doing so from  $Y$ . A report or reward calculated only from  $Y$  cannot add information. Ada gave the argument in `ada/observation_quotient.tex`, and Emmy checked it in `emmy/connection.md`.

If  $P_0 = P_1$ , the bound is  $1/2$ . That applies to a two-trial experiment made from a matched pair, or to a population whose two conditional observation laws really coincide. It does not follow for routine handovers from one matched pair. A donor with extra information about object pose could also send an informative report; that report would enlarge the receiver’s information.

### A limit from what a reward can measure

Ada also considered a finite set of feasible physical actions and a measurement that assigns each action an observed outcome. A reward may depend on that outcome, but cannot distinguish actions with the same outcome. For each private signal, Ada compared the desired action with possible deviations. These comparisons form a graph of observed outcomes. An observation based reward supports the desired actions exactly when every cycle in this graph has a nonpositive total gain from deviation. Summing the reward inequalities around a cycle proves one direction; assigning rewards from maximum path gains proves the other. The full statement and proof are in `ada/observation_quotient.tex`.

The immediate case is simple. If an actor prefers a different feasible physical action that produces the same measured outcome, the reward is identical for both actions. It cannot deter that deviation. This result assumes the actor can choose between the physical actions and that their measured outcomes are fixed. P establishes neither a strategic donor nor such a hidden deviation on its hardware.

### Q's oversight menu under a changed signal

In Q's oversight example, the state  $\theta$  is uniform on  $[0, 1]$ . A strong actor observes it exactly and has a preference bias  $b$ ; a weak monitor with bias  $w$  chooses a reward from a menu. A false report of  $b$  can drive the actor to an extreme action when the shifted state crosses an endpoint. That threat makes truthful choice of the reward best for the monitor under Q's assumptions [2].

Emmy changed the actor's observation to a signal saying only whether  $\theta$  exceeds  $1/2$ . The actor's best estimate of  $\theta$  is then either  $1/4$  or  $3/4$ . A small false report shifts its action while keeping it inside  $[0, 1]$ , so Q's extreme action threat does not arise. For an actor bias of  $1/2$ , a monitor bias of  $1/10$ , and an allowed report of  $9/20$ , the monitor's expected squared loss falls from  $37/1200$  under truth to  $7/300$  under the false report. Emmy's calculation is in `emmy/connection.md`; Ada checked the values. This shows a failure of Q's *unchanged* menu after changing its signal assumption. It does not contradict Q's result.

The peers also checked a limited repair. Suppose a signal reveals  $\theta$  with probability  $p > 0$ , and otherwise gives fresh uniform noise. Its posterior mean is  $(1 - p)/2 + ps$ , where  $s$  is the observed signal value. After rescaling actions, biases and rewards, Q's menu argument applies to this signal and implements that posterior mean. The remaining expected squared error from incomplete information is  $(1 - p^2)/12$ . This channel was introduced for analysis; P gives no evidence that handover observations follow it.

## 4 Objections and remaining uncertainty

The peers objected to treating these calculations as an established result about the paired systems. P already points to pose aware handover work and proposes a sensor test. Q already recognises that rewards depend on what can be measured. The finite graph condition is a general difference constraint argument. The oversight calculation changes Q's exact observation premise, while its repair relies on an invented signal channel. An adjacent haptic pose estimation study was recorded during the prior work search, but was not used to prove a claim or establish novelty [1].

The cross checks settled two logical points. The original oversight menu can fail under a coarse signal, yet a menu adapted to a particular signal can succeed. Likewise, identical histories in a matched pair do not imply identical observation distributions in normal operation. What remains open is physical: whether different object poses occur under matched histories, whether they demand incompatible receiving actions, and whether an actor can choose hidden physical deviations. A common safe grasp or negligible overlap would weaken the proposed connection.

## 5 Why the thread stopped

The verifier paused the thread because P does not show matched observation histories with different object poses or show that those poses require incompatible receiving actions. The information bound is therefore conditional. The oversight example changes Q's signal assumption and establishes no handover result. The smallest step to restart the thread is to hold the donor wrist fixed, externally measure a marked object's pose in occluded trials, and identify trials with matched encoder and action histories. The receiving actions for any pose distinct matches must then be tested for incompatibility. Only those measurements would show whether the mathematical limits bear on this handover.

## References

- [1] Haptic pose estimation study. arXiv:2207.02843. URL: <https://arxiv.org/abs/2207.02843>.
- [2] Dirk Bergemann, Andrew Koh, and Stephen Morris. Mechanism design for alignment and control, 2026. URL: <https://arxiv.org/abs/2609.01595>, arXiv:2609.01595.
- [3] Pathfinder research campaign. When a donor wrist cannot locate a rotating held object, 2026. Local manuscript, pathfinder-local-S5.